

❖ Generations of Computers

Generation in computer terminology is a change in technology a computer is/was being used. Initially, the generation term was used to distinguish between varying hardware technologies. Nowadays, generation includes both hardware and software, which together make up an entire computer system.

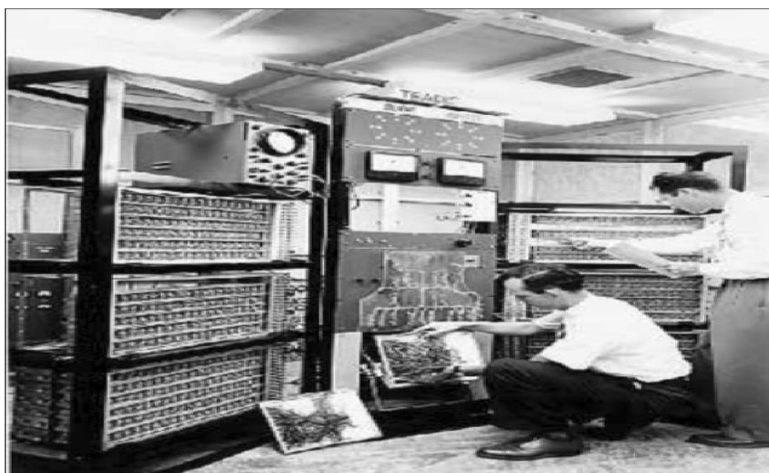
There are five computer generations known till date. Each generation has been discussed in detail along with their time period and characteristics. In the following table, approximate dates against each generation has been mentioned, which are normally accepted. Following are the main five generations of computers.

Sl. No.	Generation & Description
1	<u>First Generation</u> The period of first generation: 1946-1959. Vacuum tube based.
2	<u>Second Generation</u> The period of second generation: 1959-1965. Transistor based.
3	<u>Third Generation</u> The period of third generation: 1965-1971. Integrated Circuit based.
4	<u>Fourth Generation</u> The period of fourth generation: 1971-1980. VLSI microprocessor based.
5	<u>Fifth Generation</u> The period of fifth generation: 1980-onwards. ULSI microprocessor based.

❖ First Generation Computers

The period of first generation was from 1946-1959. The computers of first generation used vacuum tubes as the basic components for memory and circuitry for CPU (Central Processing Unit). These tubes, like electric bulbs, produced a lot of heat and the installations used to fuse frequently. Therefore, they were very expensive and only large organizations were able to afford it.

In this generation, mainly batch processing operating system was used. Punch cards, paper tape, and magnetic tape was used as input and output devices. The computers in this generation used machine code as the programming language.



The main features of the first generation are:

- Vacuum tube technology
- Unreliable
- Supported machine language only
- Very costly
- Generates lot of heat
- Slow input and output devices
- Huge size
- Need of AC
- Non-portable
- Consumes lot of electricity

Some computers of this generation were:

- ENIAC
- EDVAC
- UNIVAC
- IBM-701
- IBM-750

❖ Second Generation Computers

The period of second generation was from 1959 1965. In this generation, transistors were used that were cheaper, consumed less power, more compact in size, more reliable and faster than the first-generation machines made of vacuum tubes. In this generation, magnetic cores were used as the primary memory and magnetic tape and magnetic disks as secondary storage devices.

In this generation, assembly language and high-level programming languages like FORTRAN, COBOL were used. The computers used batch processing and multiprogramming operating system.



The main features of second generation are:

- Use of transistors
- Reliable in comparison to first generation computers
- Smaller size as compared to first generation computers
- Generates less heat as compared to first generation computers
- Consumed less electricity as compared to first generation computers
- Faster than first generation computers
- Still very costly
- AC required
- Supported machine and assembly languages

Some computers of this generation were:

- IBM 1620
- IBM 7094
- CDC 1604
- CDC 3600
- UNIVAC 1108

❖ Third Generation Computers

The period of third generation was from 1965 to 1971. The computers of third generation used Integrated Circuits (ICs) in place of transistors. A single IC has many transistors, resistors, and capacitors along with the associated circuitry.

The IC was invented by Jack Kilby. This development made computers smaller in size, reliable, and efficient. In this generation remote processing, time-sharing, multi-programming operating system were used. High-level languages (FORTRAN-II TO IV, COBOL, PASCAL, PL/1, BASIC, ALGOL-68 etc.) were used during this generation.



The main features of third generation are:

- IC used
- More reliable in comparison to previous two generations
- Smaller size
- Generated less heat
- Faster
- Lesser maintenance
- AC required
- Consumed lesser electricity
- Supported high-level language

Some computers of this generation were:

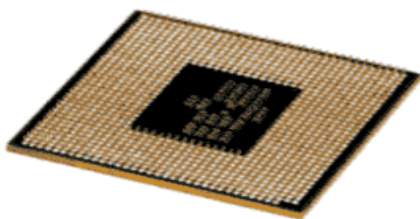
- IBM-360 series
- Honeywell-6000 series
- PDP (Personal Data Processor)
- IBM-370/168
- TDC-316

❖ Fourth Generation Computers

The period of fourth generation was from 1971-1980. Computers of fourth generation used Very Large Scale Integrated (VLSI) circuits. VLSI circuits having about 5000 transistors and other circuit elements with their associated circuits on a single chip made it possible to have microcomputers of fourth generation.

Fourth generation computers became more powerful, compact, reliable, and affordable. As a result, it gave rise to Personal Computer (PC) revolution. In this generation, time sharing, real time networks, distributed operating system were used. All the high-level languages like C, C++, DBASE etc., were used in this generation.

Fourth Generation of Computer (1971-1980)



Microprocessor



Fourth Generation Computer by IBM

The main features of fourth generation are:

- VLSI technology used
- Very cheap
- Portable and reliable
- Use of PCs
- Very small size
- Pipeline processing
- No AC required
- Concept of internet was introduced
- Great developments in the fields of networks
- Computers became easily available

Some computers of this generation were:

- DEC 10
- STAR 1000
- PDP 11
- CRAY-1(Super Computer)
- CRAY-X-MP(Super Computer)

❖ Fifth Generation Computers

The period of fifth generation is 1980-till date. In the fifth generation, VLSI technology became ULSI (Ultra Large Scale Integration) technology, resulting in the production of microprocessor chips having ten million electronic components.

This generation is based on parallel processing hardware and AI (Artificial Intelligence) software. AI is an emerging branch in computer science, which interprets the means and method of making computers think like human beings. All the high-level languages like C and C++, Java, .Net etc., are used in this generation.



The main features of fifth generation are:

- ULSI technology
- Development of true artificial intelligence
- Development of Natural language processing
- Advancement in Parallel Processing
- Advancement in Superconductor technology
- More user-friendly interfaces with multimedia features
- Availability of very powerful and compact computers at cheaper rates **Some**

computer types of this generation are:

- Desktop
- Laptop
- Notebook
- Ultrabook
- Chromebook

❖ Digital Computer

A **digital computer** is an electronic machine designed to process information using discrete values, mainly *binary digits* (0 and 1). Unlike analog computers that work with continuous signals, a digital computer handles data in exact steps, making it highly accurate, reliable, and suitable for modern computing tasks. Today's laptops, desktops, mobile devices, and embedded systems are all advanced forms of digital computers. Their power comes from the ability to perform millions of calculations per second using digital circuits built from transistors, logic gates, and microprocessors.



❖ Mini Computer

It is a midsize multi-processing system capable of supporting up to 250 users simultaneously. A **mini computer** is a mid-sized computing device that lies between a microcomputer (like a desktop or laptop) and a large-scale mainframe system. Mini computers were originally introduced to provide powerful processing capabilities at a lower cost compared to mainframes.



❖ Micro Computer

A **microcomputer** is a small, affordable, and personal computer designed for individual use. It uses a **microprocessor** as its central processing unit (CPU). Microcomputers are widely used at home, in schools, and in offices.

Examples of Microcomputers

- Desktop computers
- Laptop computers
- Tablets
- Smartphones
- Notebooks & Chromebooks



❖ Mainframe Computer

Mainframe is very large in size and is an expensive computer capable of supporting hundreds or even thousands of users simultaneously. Mainframe executes many programs concurrently and supports many simultaneous execution of programs.



❖ Super Hybrid Computer

Supercomputers are one of the fastest computers currently available.

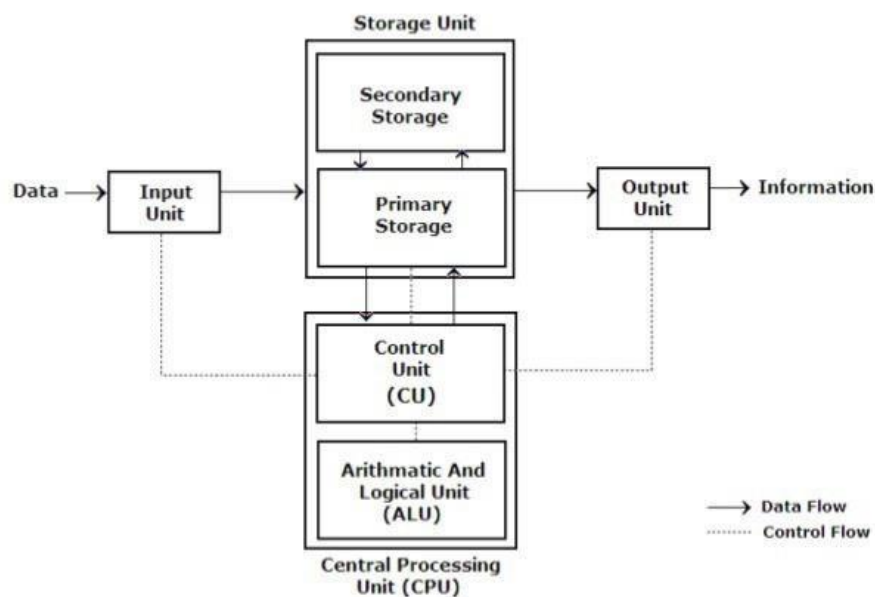
Supercomputers are very expensive and are employed for specialized applications that require immense amount of mathematical calculations (number crunching). For example, weather forecasting, scientific simulations, (animated) graphics, fluid dynamic calculations, nuclear energy research, electronic design, and analysis of geological data (e.g. in petrochemical prospecting).



❖ Block Diagram of Digital Computer

A digital computer is considered to be a calculating device that can perform arithmetic Operations at enormous speed. It is defined as a device that operates upon Information/data. To be able to process data the computer is made of various functional Units to perform its specified task.

Block diagram of computer



🔧 Arithmetic Logical Unit:

All calculations are performed in the Arithmetic Logic Unit (ALU) of the computer. It also does comparison and takes decision. The ALU can perform basic operations such as addition, subtraction, multiplication, division, etc and does logic operations viz, >, <, =, 'etc. Whenever calculations are required, the control unit transfers the data from storage unit to ALU once the computations are done, the results are transferred to the storage unit by the control unit and then it is send to the output unit for displaying results.

🔧 Control Unit:

It controls all other units in the computer. The control unit instructs the input unit, where to store the data after receiving it from the user. It controls the flow of data and instructions from the storage unit to ALU. It also controls the flow of results from the ALU to the storage unit. The control unit is generally referred as the central nervous system of the computer that control and synchronizes its working.

Central Processing Unit:

The Control Unit (CU) and Arithmetic Logic Unit (ALU) of the computer are together known as the Central Processing Unit (CPU). The CPU is like brain performs the following functions:

- It performs all calculations.
- It takes all decisions.
- It controls all units of the computer

❖ Input Devices:

Computers need to receive data and instruction in order to solve any problem. Therefore, we need to input the data and instructions into the computers. The input unit consists of one or more input devices. Keyboard is the one of the most commonly used input device. Other commonly used input devices are the Mouse, Image Scanner, Microphone etc. All the input devices perform the following functions.

- Accept the data and instructions from the outside world.
- Convert it to a form that the computer can understand.
- Supply the converted data to the computer system for further processing.

1. Mouse

A **mouse** is the most widely used pointing device with a GUI environment on personal computers. A **mechanical mouse** has a rubber ball on its underside to detect movement of the mouse. An **optical mouse** uses devices that emit and sense light to detect the mouse's movement. A **cordless mouse** (either mechanical or optical) transmits data using wireless technology such as radio waves or infrared light waves.

Advantages of using a mouse include

- _ A mouse is user-friendly for computer beginners.
- _ A mouse is easy and convenient to use with a graphical user interface.
- _ Using a mouse to select items or move to a particular position on the screen is faster than using a keyboard.

Disadvantages of using a mouse include

- _ It is not easy and convenient to input text with a mouse.
- _ Issuing commands by using a mouse is slower than by using a keyboard.
- _ A mouse is not accurate enough for drawings that require high precision.
- _ A mouse usually requires a flat surface to operate.



2. Image Scanner

A **scanner** is a light-sensing input device that converts printed text and graphics into a digital form that can be further processed by the computer.

Two popular types of scanners are:

A **flatbed scanner** works like a copy machine except that it creates a file of the document rather than a paper copy.



A **handheld scanner** can be manually passed over the image to be scanned. A portable device you can use to copy an image from paper to your computer. Hand-held scanners are ideal for capturing small images, such as signatures and logos. A hand-held scanner is smaller, less expensive, and more portable than a flatbed scanner.

Scanners are mainly used for **image processing**, which consists of capturing, storing, analysing, displaying, printing, and manipulating images. Many scanners also include optical character recognition (OCR) software, which can convert an image of text into a text file that can be edited.

Advantages of a scanner include

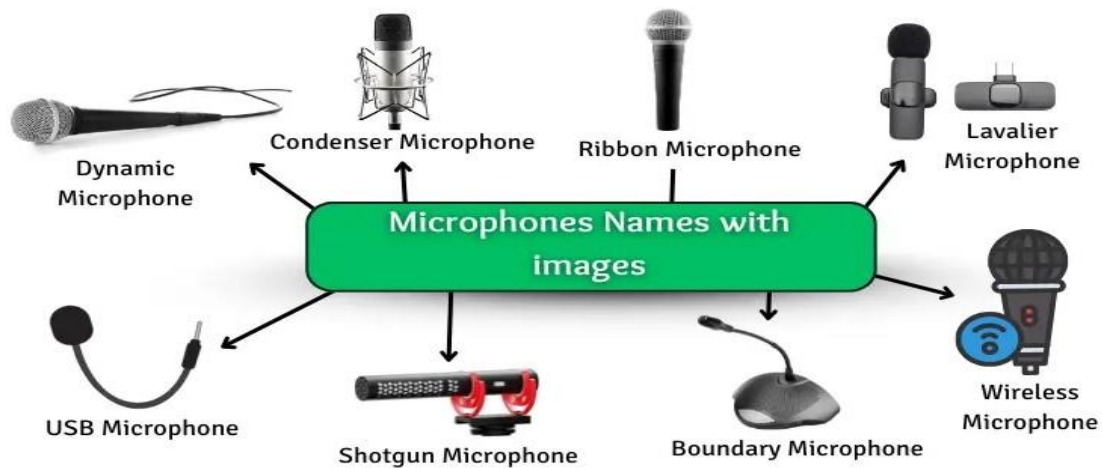
- _ A scanner with appropriate OCR software can work as an OCR system.

Disadvantages of a scanner include

- _ A scanner is best for two-dimensional objects only, difficult to scan 3-D objects
- _ Scanned images usually take up a lot of storage space.

3. Microphone

A **microphone** is an **input device** used to **enter sound or voice data** into a computer or electronic system. It converts **sound waves into electrical or digital signals** that a computer can understand and process. Microphones are widely used in communication, recording, entertainment, education, and security systems.



Working Principle of a Microphone

The basic working principle of a microphone is the **conversion of sound energy into electrical energy**.

1. When a person speaks or a sound is produced, **sound waves** travel through the air.
2. These sound waves strike a **thin diaphragm** inside the microphone.
3. The diaphragm vibrates according to the sound waves.
4. These vibrations are converted into **electrical signals**.
5. The computer or recording device processes these signals and stores or transmits them as digital audio.

Types of Microphones

1. Dynamic Microphone

Description:

Dynamic microphones use electromagnetic induction to convert sound into electrical signals. They are strong and durable.

Examples:

- Handheld stage microphones
- Public announcement (PA) systems

Uses:

- Live performances
- Public speaking
- Outdoor events

2. Condenser Microphone

Description:

Condenser microphones use a capacitor to convert sound into electrical signals. They are more sensitive and provide high-quality sound.

Examples:

- Studio microphones
- Podcast microphones

Uses:

- Music recording
- Studio production
- Broadcasting

3. USB Microphone

Description:

USB microphones connect directly to a computer through a USB port and do not require extra audio interfaces.

Examples:

- Podcasting microphones
- Online teaching microphones

Uses:

- Online classes
- Video conferencing
- Content creation

4. Wireless Microphone

Description:

Wireless microphones transmit sound signals without using physical cables.

Examples:

- Collar microphones
- Handheld wireless microphones

Uses:

- Stage performances
- Television shows
- Presentations

5. Built-in Microphone

Description:

Built-in microphones are already installed in devices such as laptops, smartphones, and tablets.

Examples:

- Laptop microphone
- Mobile phone microphone

Uses:

- Video calls
- Voice notes
- Online meetings

Uses of Microphone

Microphones are used in many fields:

- Voice recording
- Video conferencing and online meetings
- Speech recognition systems
- Online teaching and learning
- Music and studio recording
- Voice-controlled systems (AI assistants)

Advantages of Microphone

- Easy to use
- Allows hands-free input
- Enables real-time communication
- Essential for multimedia applications

Disadvantages of Microphone

- Can pick up background noise
- Quality depends on the environment
- Requires proper setup for clear sound

❖ Output Devices:

The output unit of a computer provides the information and results of a computation to Outside world. Printers, Visual Display Unit (VDU) are the commonly used output devices. Other commonly used output devices are Speaker, Headphone, Projector, Monitors, Graphic Plotter, Printer etc.

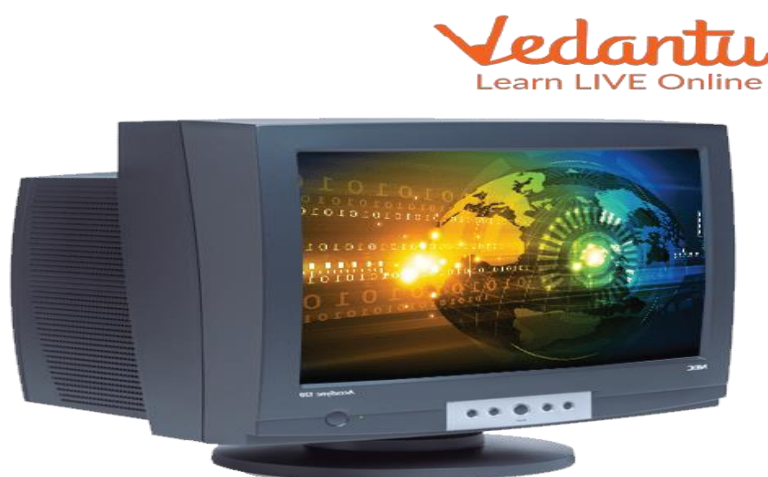
1. Monitors

Monitors, commonly called as Visual Display Unit *VDU*, are the main output device of a computer. It forms images from tiny dots, called pixels that are arranged in a rectangular form. The sharpness of the image depends upon the number of pixels.

There are two kinds of viewing screen used for monitors.

Cathode-Ray Tube *CRT*

Flat- Panel Display



Cathode-Ray Tube *CRT* Monitor

The CRT display is made up of small picture elements called pixels. The smaller the pixels, the better the image clarity, or resolution. It takes more than one illuminated pixel to form whole character, such as the letter 'e' in the word help.

A finite number of characters can be displayed on a screen at once. The screen can be divided into a series of character boxes - fixed location on the screen where a standard character can be placed. Most screens are capable of displaying 80 characters of data horizontally and 25 lines vertically. There are some disadvantages of CRT:

Large in Size

High power consumption

Flat-Panel Display Monitor

The flat-panel display refers to a class of video devices that have reduced volume, weight and power requirement in comparison to the CRT. You can hang them on walls or wear them on your wrists. Current uses of flat-panel displays include calculators, video games, monitors, laptop computer, graphics display.

The flat-panel display is divided into two categories:

Emissive Displays - The emissive displays are devices that convert electrical energy into light. Example are plasma panel and LED *Light – Emitting Diodes*.

Non-Emissive Displays - The Non-emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns. Example is LCD *Liquid – Crystal Device*



Printers

Printer is an output device, which is used to print information on paper.

There are two types of printers:

- Impact Printers

- Non-Impact Printers

Impact Printers

The impact printers print the characters by striking them on the ribbon which is then pressed on the paper.

Characteristics of Impact Printers are the following:

- Very low consumable costs

- Very noisy

- Useful for bulk printing due to low cost

- There is physical contact with the paper to produce an image

These printers are of two types

- Character printers

- Line printers

Character Printers

Character printers are the printers which print one character at a time.

These are further divided into two types:

Dot Matrix Printer *DMP*

Daisy Wheel

Dot Matrix Printer

In the market one of the most popular printers is Dot Matrix Printer. These printers are popular because of their ease of printing and economical price. Each character printed is in form of pattern of dots and head consists of a Matrix of Pins of size 5×7 , 7×9 , 9×7 or 9×9 which come out to form a character that is why it is called Dot Matrix Printer.

Advantages

Inexpensive

Widely Used

Other language characters can be printed

Disadvantages

Slow Speed

Poor Quality



Daisy Wheel

Head is lying on a wheel and pins corresponding to characters are like petals of Daisy *flowername* that is why it is called Daisy Wheel Printer. These printers are generally used for word-processing in offices which require a few letters to be sent here and there with very nice quality.

Advantages

More reliable than DMP

Better quality

The fonts of character can be easily changed

Disadvantages

Slower than DMP

Noisy

More expensive than DMP



Line Printers

Line printers are the printers which print one line at a time.

These are of further two types

- Drum Printer

- Chain Printer

Drum Printer

This printer is like a drum in shape so it is called drum printer. The surface of drum is divided into number of tracks. Total tracks are equal to size of paper i.e. for a paper width of 132 characters, drum will have 132 tracks. A character set is embossed on track. The different character sets available in the market are 48 character set, 64 and 96 characters set. One rotation of drum prints one line. Drum printers are fast in speed and can print 300 to 2000 lines per minute.

Advantages

- Very high speed

Disadvantages

- Very expensive

- Characters fonts cannot be changed

Chain Printer

In this printer, chain of character sets are used so it is called Chain Printer. A standard character set may have 48, 64, or 96 characters.

Advantages

- Character fonts can easily be changed.

- Different languages can be used with the same printer.

Disadvantages

- Noisy

Non-impact Printers

Non-impact printers print the characters without using ribbon. These printers print a complete page at a time so they are also called as Page Printers.

These printers are of two types

- Laser Printers

- Inkjet Printers

Characteristics of Non-impact Printers

- Faster than impact printers.

- They are not noisy.

- High quality.

- Support many fonts and different character size.

Laser Printers

These are non-impact page printers. They use laser lights to produce the dots needed to form the characters to be printed on a page.

Advantages

- Very high speed

- Very high quality output

- Give good graphics quality

- Support many fonts and different character size



Disadvantages

- Expensive.

- Cannot be used to produce multiple copies of a document in a single printing.

Inkjet Printers

Inkjet printers are non-impact character printers based on a relatively new technology. They print characters by spraying small drops of ink onto paper. Inkjet printers produce high quality output with presentable features.

They make less noise because no hammering is done and these have many styles of printing modes available. Colour printing is also possible. Some models of Inkjet printers can produce multiple copies of printing also.

Advantages

- High quality printing

- More reliable

Disadvantages

- Expensive as cost per page is high

- Slow as compared to laser printer



Storage Devices:

The storage unit of the computer holds data and instructions that are entered through the input unit, before they are processed. It preserves the intermediate and final results before these are sent to the output devices. It also saves the data for the later use. The various storage devices of a computer system are divided into two categories.

A) Primary Storage: Stores and provides very fast. This memory is generally used to hold the program being currently executed in the computer, the data being received from the input unit, the intermediate and final results of the program. The primary memory is temporary in nature. The data is lost, when the computer is switched off. In order to store the data permanently, the data has to be transferred to the secondary memory. The cost of the primary storage is more compared to the secondary storage. Therefore, most computers have limited primary storage capacity.

B) Secondary Storage: Secondary storage is used like an archive. It stores several programs, documents, data bases etc. The programs that you run on the computer are first transferred to the primary memory before it is actually run. Whenever the results are saved, again they get stored in the secondary memory. The secondary memory is slower and cheaper than the primary memory. Some of the commonly used secondary memory devices are Hard disk, CD, etc.

❖ Difference between Primary Storage and Secondary Storage

Basis	Primary Storage (Main Memory)	Secondary Storage (Auxiliary Memory)
Definition	Primary storage is the main memory of the computer where data and instructions are stored temporarily during processing.	Secondary storage is used to store data permanently for future use.
Volatility	Volatile in nature (data is lost when power is off).	Non-volatile (data remains stored even when power is off).
Speed	Very fast access speed.	Slower than primary storage.
Capacity	Limited storage capacity.	Very large storage capacity.
Cost	More expensive per GB.	Cheaper compared to primary storage.
CPU Access	Directly accessed by the CPU.	Not directly accessed by the CPU.
Purpose	Used for running programs and processing data.	Used for long-term storage of data and programs.
Examples	RAM, Cache Memory, ROM	Hard Disk, SSD, CD/DVD, Pen Drive